AAD3 (P25612) — Evaluation of proposed GO:0016616 annotation

Gene: AAD3 / YCR107W (*Saccharomyces cerevisiae*, NCBITaxon:559292) · **UniProt:** P25612
Focus: proposed_go_term `new-go-0016616-new` — assign **GO:0016616** *"oxidoreductase activity, acting on the CH-OH group of donors, NAD or NADP as acceptor"* (ISS, ref P 10572264) as a generalized replacement for the over-specific "aryl-alcohol dehydrogenase (NADP+) activity".

Executive Judgment

Verdict: Weakly supported / partially supported — leaning over-annotated.

The *structural* premise of the seed is correct: AAD3 is a genuine, full-length (363 aa) member of the aldo/keto reductase (AKR) superfamily (Pfam PF00248; CDD cd19147, the *Phanerochaete chrysosporium* AAD subfamily; SSF51430 NAD(P)-linked oxidoreductase), and GO:0016616 is the correct hierarchical **parent** of "aryl-alcohol dehydrogenase (NADP+) activity". So the *move* from an over-specific child term up to a broad parent is, in isolation, a defensible conservative curation action, and removing the over-specific term is clearly warranted.

However, the underlying assertion — that AAD3 has any active NADP(H)-dependent CH-OH oxidoreductase activity — is contradicted by direct primary evidence. Yang et al. 2018 (P 29079624) show that AAD3 carries a missense mutation at the **catalytically essential tyrosine (Tyr73)** and that *repairing that residue still did not yield a functional enzyme*, i.e. AAD3 is a pseudogenizing AKR with more than one inactivating lesion. Only paralogs Aad4 and Aad14 are functional.

I independently confirmed and refined the lesion. Global alignment shows AAD3 is 81.6% identical to functional Aad4 yet has **Cys** where Aad4 retains the catalytic **Tyr** (Tyr73→Cys). Motif-anchored mapping of the full **AKR catalytic tetrad** shows AAD3 retains **3 of 4** catalytic residues — **Asp68, Lys100, His148 (all present)** — but has lost only the **general-acid/base tyrosine (Tyr73→Cys)**, whereas both functional paralogs Aad4 and Aad14 carry the complete Asp/Tyr/Lys/His tetrad. Thus the AKR *scaffold and cofactor architecture are intact* (which is exactly why fold-based predictors flag AAD3), but the single catalytic residue that drives the hydride/proton chemistry is absent — and because its repair alone does not restore activity, further extra-tetrad lesions (substrate-binding loops, NADP contacts, or folding) must also contribute. Delneri et al. 1999 (P 10572264) found no aromatic-aldehyde phenotype even in a septuple aad-deletion strain.

Because a molecular-function term — even a broad one — asserts that the gene product *enables* the catalysis, GO:0016616 over-commits for AAD3. The most responsible action is to **remove the over-specific term and NOT assert an active catalytic MF**, or to keep at most a fold-based prediction that is explicitly flagged (ISS/ISM) against the experimental evidence of non-functionality.

Evidence Matrix

Citation	Evidence type	Direction	Claim tested	Key finding	Context	Confide limitati
UniProt **P25612**	database / seq record	supports (fold only)	AAD3 is a full-length AKR	363 aa, EC 1.1.1.-, "aldo/keto reductase family, AKR2 subfamily"; labelled **"Putative"**	*S. cerevisiae* record	High fo no activ demons
InterPro/ Pfam **PF00248**, CDD **cd19147**, SSF51430, PANTHER PTHR43364	computational	supports (fold only)	AKR fold + NADP-linked oxidoreductase domain	IPR023210 NADP-dependent oxidoreductase domain; cd19147 = AAD subfamily	domain analysis	High; de call ≠/ac
Own global alignment AAD3 vs **Aad4 (Q07747)**	computational	qualifies / refutes	AAD3 retains catalytic machinery?	81.6% identity (261/320); **Aad4 has Tyr where AAD3 has Cys73**; His motif (…VHWWDYM…) conserved	in-silico Needleman–Wunsch	High; indepe of litera
Own motif-anchored tetrad mapping (AAD3 vs Aad4/Aad14)	computational	qualifies / refutes	Which catalytic residues survive in AAD3?	AAD3 tetrad = **Asp68 ✓, Tyr73→Cys ✗, Lys100 ✓, His148** ✓ (3/4 intact); Aad4 & Aad14 = full Asp/Tyr/Lys/His	in-silico motif anchors (QNE, HWW, IDxA, IVIATK)	High; or catalyti is lost, s intact b chemist dead
Yang et al. 2018 — P 29079624	mutant / repair phenotype (direct)	**refutes**	Does AAD3 encode a functional enzyme?	Catalytic Tyr73 mutated; **repair did not restore function**; only Aad4/Aad14 functional; other	*S. cerevisiae* enzymology	High; di experin evidenc

Citation	Evidence type	Direction	Claim tested	Key finding	Context	Confide limitati
				AAD genes "undergoing pseudogenization"		
Delneri et al. 1999 — P 10572264 (the ISS reference)	mutant phenotype (LoF)	refutes / qualifies	Do AAD genes drive aldehyde metabolism?	Septuple deletant: **no phenotype** by spectrophotometry & HPLC	*S. cerevisiae* genetics	Mediun redunda could m but no A activity shown

Note:

P 10572264

(the cited ISS reference) is a gene-**disruption** paper that provides no positive biochemical support for AAD3 activity — it is a weak provenance choice for an activity-asserting MF.

GO Curation Implications (leads — require curator verification)

Term	Aspect	Recommended action	Rationale
"aryl-alcohol dehydrogenase (NADP+) activity" (over-specific)	MF	**Remove** (agree with review)	No substrate/activity demonstrated for AAD3; specific term unjustified
GO:0016616 (proposed parent)	MF	**Do not add as an asserted (`enables`) function**, OR add only as an explicitly-flagged fold prediction (ISS/ISM) with a pseudogene caveat	Hierarchically valid generalization, but asserts catalysis that primary data (Tyr73Cys + non-rescue) contradict
GO:0016491 "oxidoreductase activity"	MF	Acceptable *more conservative* fallback if any MF must be retained	Commits only to fold-level redox potential, not direction/cofactor

Bottom line for the curator: The review's instinct to *generalize away from over-specificity* is right, but GO:0016616 is still an activity assertion. Given experimental evidence that AAD3 is a non-functional, pseudogenizing AKR, the stronger and more defensible outcome is to leave AAD3 without an asserted catalytic MF (or with a clearly-labelled predicted term plus the Yang-2018 caveat), rather than to substitute one predicted activity term for another.

Mechanistic Scope

- **Immediate molecular function under test:** NADP(H)-dependent oxidoreduction of a CH-OH / carbonyl substrate by the AAD3 gene product itself.
- **Direct gene-product activity:** *Not demonstrated for AAD3.* The catalytic general-acid tyrosine is absent (Tyr73→Cys), and engineered restoration of that residue does not produce a functional enzyme — direct evidence that AAD3 cannot perform the catalysis.
- **Downstream/inferred effects:** No aryl-aldehyde detoxification phenotype attributable to the AAD family (septuple deletant unaffected). The family's ancestral role in lignin-derived aldehyde detox is retained only by Aad4/Aad14.

Fold membership (a structural fact) is being conflated with catalytic capability (a functional claim). The seed correctly establishes the former but the latter is refuted for this paralog.

Conflicts and Alternatives

- **Paralog over-annotation (primary concern):** The demonstrated NADPH-dependent aryl-aldehyde reductase activity belongs to **Aad4/Aad14**, not AAD3. Transferring a family-level function to AAD3 is exactly the kind of paralog carry-over that produces spurious annotations.
- **Direction of reaction:** Functional Aad enzymes physiologically *reduce* aldehydes (aldehyde + NADPH → alcohol). GO:0016616 describes the *oxidation* direction (CH-OH + NAD(P)+). The term is the correct EC/GO parent of "aryl-alcohol dehydrogenase (NADP+)", but it does not describe the enzymes' physiological reductase direction — another reason the term is an imperfect fit even for the functional members.
- **Pseudogenization gradient:** AAD3, AAD6, AAD10, AAD15, AAD16 are degenerate to varying degrees; AAD3 specifically is a full-length ORF whose lesions are point/missense rather than truncation — which can make automated fold-based pipelines mis-predict activity.

Knowledge Gaps

1. **No direct in-vitro assay of purified AAD3 protein.** Checked: literature (Yang 2018 tested a *repaired* AAD3, not wild-type in a substrate panel). Matters because a definitive negative enzymatic result on WT AAD3 would fully settle the MF question. Resolve by: express/purify AAD3, assay aryl-/aliphatic-aldehyde reduction with NADPH (veratraldehyde, cinnamaldehyde, vanillin).
2. **Is AAD3 expressed/translated?** Checked: not established here. Matters for whether any GO MF annotation is even appropriate vs treating as a pseudogene. Resolve by: ribosome profiling / proteomics evidence.
3. **Complete catalog of inactivating substitutions.** Checked (partly resolved here): tetrad mapping shows only the catalytic **Tyr73** is lost in AAD3 (Asp68/Lys100/His148 retained), so the non-rescuable phenotype (Yang 2018) points to lesion(s) **outside** the tetrad. Matters for confidence in "non-functional" and for whether a fold-based MF could ever be defensible.

Resolve by: map NADP-binding pocket + substrate loops in AAD3 vs Aad4/Aad14 and perform combinatorial site-directed rescue.

4. **NADP(H) binding retained?** The C-terminal cofactor loop appears intact, but binding was not measured. Resolve by: ITC/fluorescence cofactor-binding assay.

Discriminating Tests

- **Recombinant enzyme assay** of wild-type AAD3 (± the Tyr73 repair, ± additional tetrad repairs) on a defined aryl/aliphatic aldehyde panel with NADPH — directly distinguishes "inactive pseudogene" from "cryptically active oxidoreductase."
- **Ancestral-state reconstruction / full tetrad repair** (as Yang did for Aad6-16 and Aad10) to pinpoint how many lesions separate AAD3 from function.
- **Expression evidence** (RNA-seq under DTT/oxidative stress; proteomics) to decide whether AAD3 warrants any protein-level GO annotation.
- **Comparative tetrad mapping** across all seven yeast AAD paralogs to build a per-paralog functionality call for consistent curation.

Curation Leads (require curator verification)

- **Add reference**

  P 29079624

  (Yang et al. 2018) — verbatim snippet to verify: *"Repair of an AAD3 missense mutation at the catalytically essential Tyr73 residue did not result in a functional enzyme."* and *"two of these genes, AAD4 and AAD14, encode functional enzymes that reduce aliphatic and aryl-aldehydes concomitant with the oxidation of cofactor NADPH."*
- **Re-examine the ISS reference**

  P 10572264

  — it is a disruption paper reporting *no* aldehyde phenotype; verify snippet *"None of the knock-out strains revealed any mutant phenotype when tested for the degradation of aromatic aldehydes."* It does not positively support an activity MF.

- **Candidate action change:** Remove the over-specific aryl-alcohol-dehydrogenase (NADP+) MF (agree). For the replacement, prefer **no asserted catalytic MF**, or GO:0016491/ GO:0016616 only as an explicitly-flagged prediction (ISS/ISM) with a pseudogenization caveat citing

  P 29079624

  — rather than substituting GO:0016616 as an `enables` annotation.
- **Suggested question for curator:** Should AAD3 be treated as a pseudogenizing ORF (annotate fold membership + note loss of catalytic Tyr73) rather than as an active oxidoreductase?
- **Suggested experiment:** WT-AAD3 recombinant aldehyde-reductase assay with NADPH (definitive).

Provenance artifacts

- `/tmp/AAD3_evidence_matrix.csv` — evidence matrix (this analysis)
- `/tmp/AAD3_catalytic_tetrad.png` — AAD3 vs Aad4 alignment showing Tyr73→Cys catalytic-residue loss
- `/tmp/AAD3_tetrad_table.csv` / `/tmp/AAD3_tetrad_table.png` — motif-anchored AKR tetrad comparison (AAD3 vs Aad4/Aad14): 3/4 retained, only catalytic Tyr lost
- `/tmp/AAD3_GO_decision_table.csv` — GO decision table (attempted; regenerate if absent)

Computational results (UniProt fetch, InterPro domain calls, AAD3-vs-Aad4 global alignment) were executed live; the Tyr73→Cys finding is reproduced independently of the cited literature and is concordant with it.
